

IMAEN: An interpretable molecular augmentation model for drug–target interaction prediction

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ABSTRACT

Drug discovery is a crucial aspect of biomedical research, and predicting drug–target interactions (DTIs) is a vital step in this process. Graph neural networks (GNNs) have achieved remarkable performance on graph learning tasks, including DTI prediction. However, existing DTI models using GNNs are insufficient to aggregate node neighborhood information and lack interpretability. In this study, we propose an interpretable molecular augmentation model named IMAEN for drug–target interaction prediction, which employs molecular augmentation mechanism to fully aggregate the molecular node neighborhood information, especially for the nodes with fewer neighborhoods. Moreover, we design an interpretable stack convolutional encoding module to process protein sequence from the perspective of multi-scale and multi-level interpretability. Compared with the existing models, our proposed model has the best effect and achieves the best performance on four benchmark datasets. The visualization of the model and interpretation of its predictions provide valuable insights into the underlying mechanisms of drug–target interactions, which could assist researchers in narrowing the search space of binding sites and aid in the discovery of new drugs and targets. The source code and experimental datasets can be found in <https://github.com/zhangjing-dmu/IMAEN>.

1. Introduction

Since the development of new drugs is an expensive and time-consuming process, the use of FDA-approved drugs for some diseases have potentially lower overall development costs and shorter development times. In the current biomedical domain, computational drug repurposing methods have become a pressing need to accelerate drug discovery (Agamah et al., 2020; Ali, Min, Xiao-Li, & Chee-Keong, 2023). Predicting DTIs is an important procedure in drug discovery and repurposing (Zhao, Yang, Cheng, Li, & Wang, 2021), and has received significant attention from researchers due to the availability of a mass of biomedical data and knowledge. In silico methods, DTI prediction is a high-efficiency and low cost method that are widely preferred. Currently, there are three main types of DTI prediction methods: structure-based methods, ligand-based methods, and machine-learning-based methods.

In the early stage, most DTI prediction methods were based on the three-dimensional structure (Feixiong et al., 2012), which allowed researchers to probe potential binding sites. Recently, Shaikh, Sharma, and Garg (2016) advanced protein chemometrics (PCM) for DTI prediction, their method can generate dependable negative datasets and

analyze the fingerprint-based applicability domain. The reverse molecular docking method is then used to quantitatively predict the molecular recognition interaction between drug and target pairs. Molecular dynamics simulation is another widely used method that can provide detailed information on the movement of individual particles over time. Nevertheless, structure-based methods heavily rely on protein's three-dimensional structure, making them ill-suited for proteins with unknown three-dimensional structures.

Ligand-based drug design is a well-established computational approach in the field of drug discovery and development that utilizes the structural information of known active small molecules to predict the activity of novel compounds or improve the efficacy of existing compounds. This approach is accomplished by developing pharmacophore models or quantitative structure–activity relationships (QSARs), which can guide the structural optimization of drug candidates. The pharmacophore model can indirectly deduce the active site's structure of the target and provide insights into the mechanism of action of the compound. Additionally, the pharmacodynamic group model can be used to screen compound databases for specific pharmacodynamic characteristics. Recently, Gonzalezdiaz et al. (2010) proposed a multitarget

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QSAR classifier for predicting DTIs with notable success. However, the ligand-based approach may not be as effective in the case of only a few known ligands bind to the target.

In recent years, machine-learning-based methods have been utilized for predicting DTIs. Although early work in this field focused on pharmacological DTI prediction, recent research has broadened its scope to include the identification of sequencing lists of molecular ligands that bind to individual G protein-coupled receptors (GPCRs). Notably, some of the most popular DTI prediction methods have included affinity similarity, such as SimBoost (He, 2016), and Kronecker regularized least square methods, like KronRLS (Cichonska et al., 2023, 2017). However, over the last few years, a plethora of machine learning-based approaches have been proposed (Ingoo & Hojung, 2018; Mohamed, Nováek, & Nounu, 2019; Monteiro, Oliveira, & Arrais, 2022; Olayan, Haitham, & Bajic, 2018; Qingjun et al., 2023; Zong, Kim, Ngo, & Harismendy, 2017). Qingjun et al. (2023) combines the strengths of feature-based and similarity-based approaches to improve prediction performance. Olayan et al. (2018) relies on DTI heterogeneous graphs, nonlinear similarity fusion methods, and feature extraction from heterogeneous images trained on random forests to predict DTIs. Additionally, Zong et al. (2017) proposed an approach for predicting drugs and targets based on similarity using topology-based similarity measures to enhance existing association discovery methods. They used the deep learning method DeepWalk to calculate similarity within a heterogeneous network, the Linked Tripartite Network (LTN), which was generated from biomedical associated datasets. However, while machine learning-based methods have shown promise in predicting DTIs, they often lack interpretability.

Predicting DTIs enables the identification of new drug targets and the design of effective drugs. Drug–drug interactions also play a crucial role in drug repositioning. Li, Hu, Wang, Li, and You (2022) propose a novel approach, MNMDCDA, which combines high-order graph convolutional networks and deep neural networks to predict associations between circRNAs and diseases, addressing the challenge of limited circRNA–disease associations and the time-consuming nature of wet-lab methods. Su et al. (2023) propose a new framework called KG2ECapsule, which explicitly models multi-relational drug–drug interaction (DDI) data within biomedical knowledge graphs in an end-to-end manner. This framework generates high-quality negative samples, optimizes entity representations through recursive propagation of embeddings from attention-based receptive fields, and performs non-linear transformations in specified relational spaces. As a result, it effectively addresses the challenge of multi-relational DDI prediction. In recent years, deep learning has emerged as a promising approach for DTI prediction, as it has achieved excellent performance in various fields, especially in computer vision and natural language processing (Hazra, Choudhary, & Sheetal Singh, 2021; Lecun, 2023). In the realm of DTI prediction, previous studies have overlooked the issue of obtaining more node representation from nodes with fewer neighborhoods, while also lacking interpretability of their models. To address these challenges, we propose a novel approach called the Molecular Structure Augmentation Networks and Interpretable Stack Convolutional Neural Network (IMAEN). Our method leverages molecular structure augmentation networks to capture additional graph topological information and transmit drug compound information more efficiently. Moreover, we incorporate the interpretable stack convolutional neural network module to effectively process target sequences. Our experimental results demonstrate that our proposed model outperforms existing methods in affinity prediction tasks when trained on four benchmark datasets, including Davis (Davis et al., 2011a), KIBA (Tang, Szwajda, Shakyawar, Tao, & Aittokallio, 2014), DTC (Metz et al., 2023), and Metz (Jing, Tanoli, Ravikumar, Alam, & Aittokallio, 2018). The main contributions of this study are threefold:

1. We propose an interpretable molecular augmentation encoding networks named IMAEN for drug–target interaction prediction, which employs molecular augmentation mechanism to

fully aggregate the molecular node neighborhood information, especially for node with fewer neighborhoods.

2. We design an interpretable stack convolutional encoding module to process protein sequence from the perspective of multi-scale and multi-level interpretably.
3. Extensive experiments have shown that the proposed method achieves state-of-the-art performance on four benchmark datasets. The visualizations and case studies analysis demonstrate that the molecular augmentation and interpretability mechanism can aid in DTI tasks.

2. Related work

Several early deep learning methods proposed for drug repurposing include Chu, Liu, and Zhang (2022), Fu, Huang, Liu, Qiu, and Zhang (2021), Hakime, Arzucan, and Elif (2018), Lin et al. (2020), Majumdar et al. (2021), Öztürk, Ozkirimli, and Özgür (2019), Shin, Park, Kang, and Ho (2019), Tsubaki, Tomii, and Sese (2019), Wang, Huang, Xiong, and Zhang (2022), Zhao, Wang, Pang, Liu, and Zhang (2020), Zhao, Xiao, Yang, Li, and Wang (2023). Currently, several deep learning models, including MolTrans (Lin et al., 2020), DeepGLSTM (Mukherjee, Ghosh, & Basuchowdhuri, 2022), IIFDTI (Cheng, Zhao, Li, & Wang, 2022), AttentionSiteDTI (Yazdani-Jahromi et al., 2022), and HyperattentionDTI (Zhao, Zhao, Zheng, & Wang, 2022), have been proposed for DTI prediction. Lin et al. (2020) proposed a molecular interaction converter (MolTrans) that utilizes substructure pattern mining and interaction modeling to extract and capture semantic relationships between substructures extracted from large amounts of unlabeled biomedical data. Mukherjee et al. (2022) proposed a new architecture, DeepGLSTM, based on graph convolutional networks and LSTM, which effectively solves the problem of obtaining amino acid sequence context information. Cheng et al. (2022) proposed IIFDTI that considers the independent characteristics of drug/target pairs and the interaction characteristics of their substructures. Fu et al. (2021) introduced MVGCN, presenting an innovative framework for link prediction in biomedical bipartite networks. Their approach effectively utilizes multi-view heterogeneous networks and aggregation strategies to enhance performance. Chu et al. (2022) established a hierarchical graph learning architecture to integrate coarse- and fine-level information from an affinity graph and drug/target molecule graphs. Additionally, a similarity-based representation inference method was designed to address the cold start scenario. Zhao et al. (2022) proposed an end-to-end bio-inspired model for predicting DTIs based on CNN and attention mechanisms, using deep convolutional neural networks to learn eigenmatrices of drugs and proteins. Wang et al. (2022) introduced a model named HampDTI, utilizing a novel HN-based approach with attentive meta-path extraction, achieving improved DTI prediction performance by automatically extracting valuable meta-paths and optimizing them during model training. Yazdani-Jahromi et al. (2022) addressed the problem of drug–target interaction prediction by utilizing protein binding sites and a self-attention mechanism. They introduced an interpretable graph-based deep learning prediction model called AttentionSiteDTI. This work has greatly inspired our own research.

3. Method

In this section, we will describe our proposed method IMAEN in details, which comprises four parts: the input representation module, molecular structure augmentation module, interpretable stack convolutional encoding module, and drug–target interaction prediction module. As illustrated in Fig. 1, we first learn the representations of drugs and proteins. These representations are then passed through the molecular structure augmentation module and the interpretable stack convolutional neural network module, respectively. The output representations of the two modules are concatenated and input into the drug–target interaction prediction module to obtain the affinity score. In the following subsections, we describe of each of the four modules in detail.

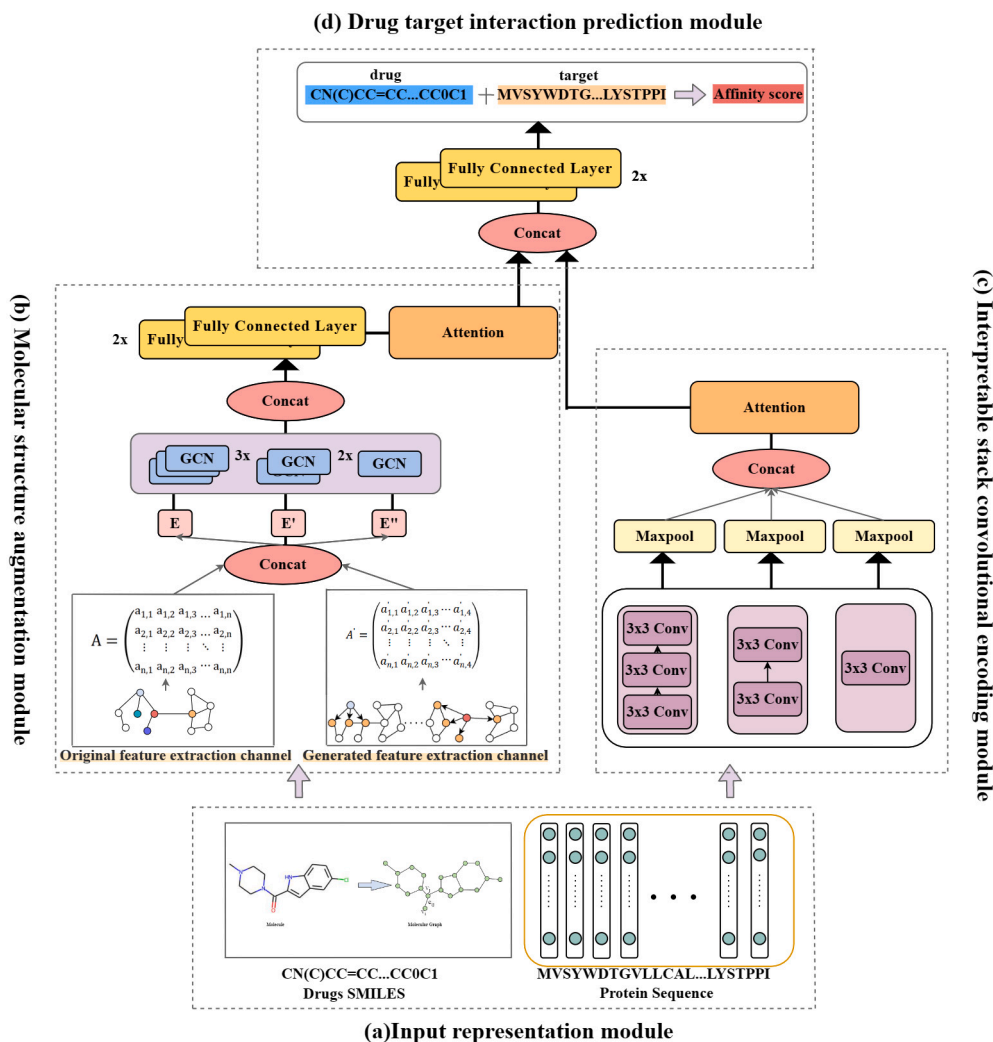


Fig. 1. Overview of the proposed IMAEN. The model is composed of four parts, namely input representation module(a), molecular structure augmentation module(b), interpretable stack convolutional encoding module(c) and drug target interaction prediction module.

3.1. Input representation module

In this study, we have utilized Simplified Molecular Input Line Entry System (SMILES) (David, 1988) to represent each drug compound. We utilize the open-source chemical information software RDKit (Landrum et al., 2016) to convert SMILES data into the corresponding molecular graph representation, which are with vertex (or node) features and adjacency matrix(i.e. adjacency representation $A \in \mathbb{R}^{N \times N}$), as illustrated in the left graph of Fig. 2. Finally, we obtain the topological information of the drug compound, which serves as input for the molecular structure augmentation module(MSAM).

The protein sequence is a string of ASCII characters. To facilitate training, we set the maximum length of the protein sequence to 1000 to cover at least 80% of the proteins. For the input representation of the target, we use the popular unique thermal code (Sohn, Yan, Lee, & Arbor, 2023) to represent the protein sequence. Each amino acid is mapped to a letter, and the amino acid is labeled with the sequential index of the letter. The letter is transformed into a series of integer by using Char-Dict Encoding. The protein sequence embedding layer is then initialized. To transform the complex amino acid sequence into a format that can be processed by computers, we map each integer to a 128-dimensional vector by using One-Hot Encoding. The right graph in Fig. 2 illustrates this. We pass the embedding representation ($c \in \mathbb{R}^{d \times p}$) to the interpretable stack convolutional encoding module(ISCEN), which captures the dependencies between characters in a sequence of length n ($p = [c_1, c_2, \dots, c_n]$).

3.2. Molecular structure augmentation module

In this section, we introduce the Molecular Structure Enhancement Module (MSAM) depicted in Fig. 3(b). This module adopts a straightforward yet powerful mechanism to enhance molecular structures. By inputting the molecular graphs into both the original and enhanced feature extraction channels, we extract their corresponding original and enhanced features. The former utilizes a standard molecular feature extraction method, while the latter samples the neighboring information of each node in the molecular graph to extract generative features. To amplify the number of samples within a node v 's neighborhood N_v , we must grasp the distribution of node representations among its neighbors. Since this distribution correlates with the central node v , we leverage a generative model to learn it from the central node's representation, thereby bolstering the GNN's expressive capacity through these generative features.

Our approach treats the central node as the parent (represented by blue, green, red, and purple circles), and its neighbors as children (depicted as saffron circles). After gaining insight into the local neighborhood distribution, we extract features for each node in the graph based on this distribution. By merging the original and enhanced feature matrices, we construct a new matrix that is then fed into the downstream Multi-Scale GCN module. This approach effectively captures both local and global structural aspects within the molecular graph, enabling comprehensive exploration of node neighborhood information.

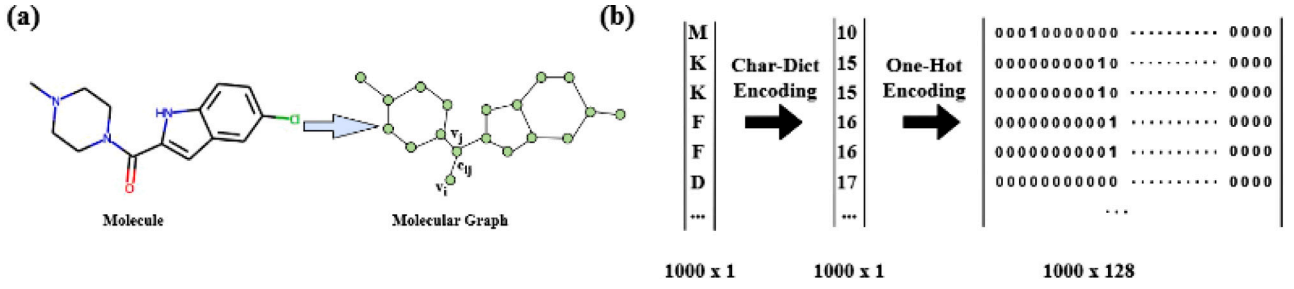


Fig. 2. The process of drug-protein input representation. (a) Molecule is converted to molecule graph. (b) The process of protein sequence embedding.

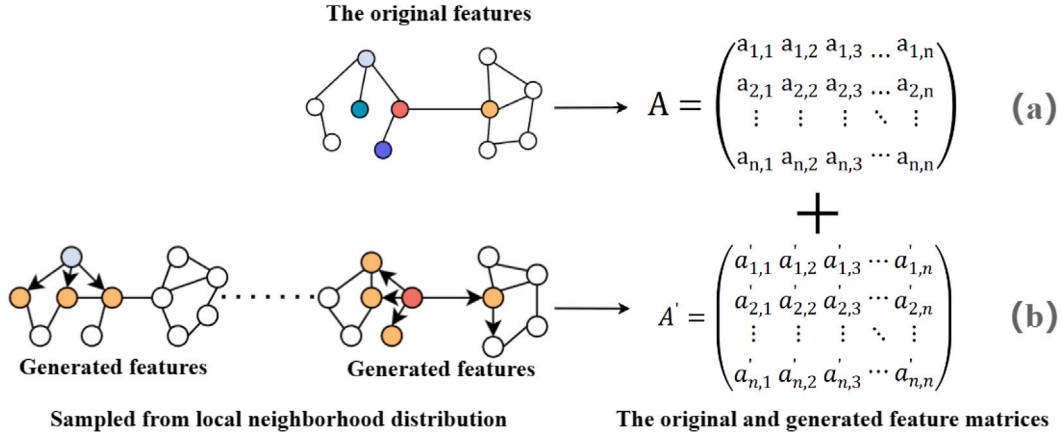


Fig. 3. Two channels for molecular feature extraction. (a) The original feature extraction channel. (b) Generated feature extraction channel. Then we concatenate the original feature matrix and the generated feature matrix to form a new feature matrix, which is then input into the downstream multi-scale GCN module.

According to the conditional variational autoencoder (Kingma & Ba, 2014; Sohn et al., 2023), A_v is set to a condition, and $A_u (u \in N_v)$ is related to A_v . Following Sohn et al. (2023), with the help of the prior distribution $p_\theta(z|A_v)$, we can get the latent variable z . We use the neighboring pairs $(A_v, A_u, u \in N_v)$ as input to maximize the evidence lower bound to get the generated features matrices A' , by using Eq. (1).

$$\mathcal{L}(A_u, A_v; \theta, \phi) = -K N(q_\phi(z|A_u, A_v) \parallel p_\theta(z|A_v)) + \frac{1}{N} \sum_{i=1}^N \log p_\theta(A_u|A_v, z^{(i)}) \quad (1)$$

where $u \in N_v$, $z^{(i)} = g_\phi(A_v, A_u, \epsilon^{(i)})$, $\epsilon^{(i)} \sim N(0, I)$, $q_\phi()$ and $p_\theta()$ represent the prior distribution, N is the number of neighbors of node v . In the generation stage, we use node feature A_v as the condition and sample potential variables $z^{(i)} \sim N(0, I)$ as the input of decoding. Then we can get the generated feature vector A' associated with node v , namely the generated feature matrices.

For each drug compound, following Xu, Hu, Leskovec, and Jegelka (2018), we figure up a propagation rule as mentioned via taking the adjacency representation(Q), the original node feature matrix(X) and the generated node feature matrix(A') generated from the local augmentation module as inputs into the first block. We input the final feature matrices E into the downstream multi-scale GCN module.

$$E = (X \parallel A') \quad (2)$$

$$Q_{norm} = D^{-\frac{1}{2}} Q D^{-\frac{1}{2}} \quad (3)$$

where $D \in \mathbb{R}^{N \times N}$ is the degree matrix representation of Q .

$$H_e^{(i-1)} = \sigma(Q_{norm} H_e^{(i-1)} W^{(i-1)}) \quad (4)$$

where W is the trainable weight, $H_e^{(0)} = E$ and σ is a non-linear activation function. $H_e^{(i-1)}$ is the global representation of i th layer of the GCN module.

We construct a graph that can be expressed as power graph of power 2 if central node is pertained to all such nodes that are two hops away from it.

Similar to the first block, the normalized adjacency representation and the final representation of the second GCN block are as follows:

$$Q_{norm}^2 = D'^{-\frac{1}{2}} Q^2 D'^{-\frac{1}{2}} \quad (5)$$

where Q_{norm}^2 is squared representation of Q , D' is the degree representation of Q^2 .

$$H_e^{''i} = \sigma(Q_{norm}^2 H_e^{''(i-1)} W^{''(i-1)}) \quad (6)$$

where $H_e^{''i}$ is the global representation of the second GCN module. When $i = 0$, $H_e^{''0} = E$. In the same way, for the third block, the normalized adjacency representation and the final representation of the third GCN block are as follows:

$$Q_{norm}^3 = D''^{-\frac{1}{2}} Q^3 D''^{-\frac{1}{2}} \quad (7)$$

$$H_e^{'''i} = \sigma(Q_{norm}^3 H_e^{'''(i-1)} W^{'''(i-1)}) \quad (8)$$

where Q_{norm}^3 is squared representation of Q , D'' is the degree representation of Q^3 , $H_e^{'''i}$ is the global representation of the third GCN module. When $i = 0$, $H_e^{'''0} = E$.

Then we concatenate the output representations of mentioned above GCN module to get the final representation, as shown in Eq. (9).

$$H_e = (H_e^i \cdot H_e^{''i} \cdot H_e^{'''i}) \quad (9)$$

Finally, we input H_e into the attention layer to get the final output representation X_d by using Eq. (10).

$$X_d = \text{SOFTMAX}(\tan(H_e)) \quad (10)$$

3.3. Interpretable stack convolutional encoding module

With the TextCNN presented by Kim (2014) for text classification, CNN has gradually become a research hotspot in the field of natural language processing. In this study, we took inspiration from the drug processing process and used ISCEM which adopts interpretable attention mechanism to extract protein features on a multi-scale. Specifically, we proposed a stack convolutional neural network based on the attention mechanism, which is a three-branch convolutional network with different receptive domains.

We input the embedding representation s ($s \in d_p$, where d_p is the embedding dimension of the protein sequence) into the interpretable stack convolutional encoding module to obtain the output representation $h_i \in \mathbb{R}^d$.

$$h_i = w(m(g_1(s)) \| m(g_2(s)) \| m(g_3(s))) \quad (11)$$

where g_i is the i -th branch with 3×3 convolutional layers that maps s into a matrix $C \in \mathbb{R}^{1000 \times h}$ in which ReLU activation were added behind each convolutional layer, m represents maxpool operation, and $w \in \mathbb{R}^{d \times 3h}$ is a learnable matrix. Then input h_i to the attention layer by using Eq. (12).

$$X_i = \text{SOFTMAX}(\tan(h_i)) \quad (12)$$

The ISCEM is composed of an attention mechanism module and a Stack-CNN module, which work together to extract features from protein sequences. This model effectively combines the strengths of attention mechanism and CNN to obtain both contextual information and long dependency information from protein sequences. In the experimental part, the interpretability of the model is demonstrated through visualization of attention weights, which provide insights into which parts of the input are more important for prediction. The implementation of ISCEM is as follows: (1) Input Representation Transformation: First, the embedded representation of the protein sequence is passed through an interpretable stacked convolutional encoding module. This embedded representation can be obtained through a pre-trained embedding model. (2) Convolutional Encoding: In this module, there are three branches, each containing a series of 3×3 convolutional layers that transform the input embedded representation into a matrix. A ReLU activation function is applied after each convolutional layer. The design of these three branches allows each branch to capture different features and contextual information. (3) Pooling and Stacking: Within each branch, maxpool operations are used to pool the features after convolution, extracting key information. Subsequently, these pooled features are concatenated together to obtain an overall encoding representation that will be used in the subsequent attention layer. (4) Attention Mechanism: The encoding representation is input into the attention layer, where relevant formulas are applied to perform softmax normalization and activation function transformation. This step enhances the model's focus on different features, thereby emphasizing information that contributes more to predictions.

4. Experiments design

4.1. Datasets

This work uses the Davis (Davis et al., 2011b), KIBA (Tang, Szwajda, Shakyawar, Xu, & Aittokallio, 2014), DTC (Tang, Tanoli, Ravikumar, Alam, & Aittokallio, 2018), Metz (Metz, Johnson, Soni, & Merta, 2011) and Human datasets, and evaluates the performance of our model using Mean Squared Error (MSE), Concordance Index (CI), and r_m^2 , as detailed in Table 1. The distribution of binding affinity values was significantly biased towards $K_d = 10,000$ nM on Davis (Davis et al., 2011a), which are rarely observed interactions. Additionally, the values variance range from near 0 to high values (weak bindings). To avoid unnecessarily losses, we follow the work of our predecessors that convert k_d values to logarithmic space (pk_d). Similar to the method

Table 1

Summary of the four datasets.

Dataset	Number of compounds/drugs	Number of targets	Total number of pair samples
Davis (k_d)	68	442	30056
KIBA	2111	229	118254
DTC (pk_i)	5983	118	67894
Metz (pk_i)	1471	170	35307
Human	2726	2001	6728

Table 2

Parameter settings.

Parameter	Value
epochs	1000
learning rate	0.0005
batch size for Davis	128
dropout probability (p)	0.2
protein sequence length (n)	1000
input node features numbers (C)	78
batch size for KIBA, DTC and Metz	512
GCN block ^{1st} final hidden vector representation (M')	312
GCN block ^{2nd} final hidden vector representation (M')	156
GCN block ^{3rd} final hidden vector representation (M'')	78

used in He (2016), we transform the k_d value of the Davis dataset into log space pk_d for Davis dataset affinity prediction.

$$pk_d = -\log_{10} \left(\frac{K_d}{1e9} \right) \quad (13)$$

4.2. Evaluation metrics

In our experiments, we used the ADAM (Kingma & Ba, 2014) optimizer and all experimental setups are shown in Table 2. Owing to this is a regression problem, we use MSE, CI, r_m^2 as evaluate metrics. MSE is a common evaluation metric used to measure the difference between predicted values (P) and actual values (Y). In this case, for Davis, we use pk_d values, while for KIBA, the values are KIBA scores.

$$MSE = \frac{1}{B} \sum_{i=1}^B B(P_i - Y_i)^2 \quad (14)$$

where B indicates the number of samples.

CI evaluates the models' ranking performance:

$$CI = \frac{1}{z} \sum_{\delta_i - \delta_j} h(b_i - b_j) \quad (15)$$

where b_i is the prediction value for δ_i , b_j is the prediction value for δ_j , z is a normalization constant. $h(m)$ is the step function:

$$h(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0.5, & \text{if } x = 0 \\ 0, & \text{if } x < 0 \end{cases} \quad (16)$$

In addition, the metric r_m^2 is introduced as a measure in the proposed method. The calculation of r_m^2 is described in Eq. (17):

$$r_m^2 = r_m^2 \times (1 - \sqrt{r_m^2 - r_m^0}) \quad (17)$$

5. Results and discussion

5.1. Compare with SOTA DTI prediction method

In our study, we employed the evaluation metrics mentioned above to compare the performance of our proposed IMAEN model with several previously proposed models, including DeepGLSTM (Mukherjee et al., 2022), SOTA DeepDTA (Hakime et al., 2018), MT-DTI (Shin, Park, Kang, et al., 2019), GraphDTA (Nguyen, Le, Quinn, Nguyen, & Venkatesh, 2020) and WideDTA (Öztürk et al., 2019). We also

Table 3

Comparison results of the proposed IMAEN and baselines on the Davis and KIBA datasets.

Model	Dataset		Davis (pk_d)			KIBA		
	Proteins	Compounds	MSE	CI	r_m^2	MSE	CI	r_m^2
DeepCPI	CNN	GNN	0.293	0.867	0.607	0.211	0.852	0.657
Attention-DTI	CNN	CNN	0.245	0.887	0.657	0.193	0.882	0.738
DeepDTA	CNN	CNN	0.261	0.878	0.630	0.194	0.863	0.673
WideDTA	CNN+PDM	CNN+LMCS	0.262	0.886	0.635	0.179	0.875	0.704
KronRLS	SW	PS	0.379	0.871	0.407	0.411	0.782	0.342
SimBoost	SW	PS	0.282	0.872	0.655	0.222	0.836	0.629
DeepGLSTM	GCN	Bi-LSTM	0.232	0.895	0.680	0.133	0.897	0.792
MT-DTI	CNN	Transformer	0.245	0.887	0.665	0.193	0.882	0.738
GraphDTA	CNN	CNN	0.254	0.880	0.659	0.139	0.889	0.752
GraphDTA	CNN	GAT	0.232	0.892	0.664	0.179	0.866	0.763
GraphDTA	CNN	GIN	0.229	0.893	0.660	0.147	0.882	0.772
GraphDTA	CNN	GAT-GCN	0.245	0.881	0.642	0.139	0.891	0.780
FNN	ECFP	PSC	0.244	0.893	0.685	0.216	0.818	0.659
RF	ECFP	PSC	0.359	0.854	0.549	0.245	0.837	0.581
SVM	ECFP	PSC	0.383	0.857	0.513	0.308	0.799	0.513
IMAEN	MGCN	Att-MGCN	0.211	0.905	0.705	0.130	0.898	0.775

Table 4

Comparison results of the proposed IMAEN and baselines on the DTC and Metz datasets.

Dataset	Model		DTC (pk_d)			Metz (pk_d)		
	Proteins	Compounds	MSE	CI	r_m^2	MSE	CI	r_m^2
GraphDTA (GCN)	CNN	GCN	0.317	0.878	0.812	0.317	0.801	0.620
GraphDTA (GAT-GCN)	CNN	GAT-GCN	0.200	0.857	0.790	0.333	0.795	0.602
GraphDTA (GIN)	CNN	GIN	0.176	0.876	0.798	0.317	0.800	0.645
GraphDTA (GAT)	CNN	GAT	0.195	0.859	0.788	0.393	0.775	0.549
DeepGLSTM	GCN	Bi-LSTM	0.149	0.895	0.841	0.294	0.810	0.640
IMAEN	MGCN	Att-MGCN	0.144	0.901	0.836	0.284	0.814	0.654

compare the proposed method with five traditional PCM models: SimBoost, KronRLS, support vector machine(SVM), random forest(RF) and feedforward neural network (FNN). The hyperparameters used for the baseline models were obtained from recommended values in prior research. We performed hyperparameter search based on factors such as our computer's performance and other considerations. We employed a grid search approach to explore the optimal combinations in the hyperparameter space. For each set of hyperparameters, we trained the model on the training set and evaluated its performance on the validation set. Based on the performance assessment on the validation set, we adjusted the values of the hyperparameters. If we observed performance improvement, we iterated further in that direction; if not, we explored other combinations of hyperparameters. The training-validation set split ratio for all baseline datasets was set to 5:1. For purpose of learning a generalized model, we conducted ten cross-validations on four datasets to obtain experimental results and took their average value as the final result. In this section, we present the experimental results of IMAEN and compare it with several state-of-the-art methods, as shown in Tables 3 and 4. As shown in Table 3, for Davis, IMAEN has made a huge improvement compared with baseline. Contrasted with the competitive method, IMAEN achieves an improvement of 2.10% for MSE and 2.50% for r_m^2 . As shown in Table 4, for Metz dataset, IMAEN achieves an improvement of 1.0% for MSE and 1.4% for r_m^2 . To demonstrate the robustness of the model, we conducted a classification task on a human dataset. As shown in Table 5, our experiments yielded competitive results among multiple models.

Fig. 4 explains the binding affinity of predicted and actual value of the Davis testset, where we can observe that the data is particularly dense near the reference line.

5.2. Ablation study

In order to further sustain the IMAEN architecture, we performed ablation experiments on the proposed model, specifically by removing (i) the Local Augmentation block, (ii) the ISCEM and (iii) the Attention

Table 5

Comparison results of the proposed IMAEN and baselines on the Human datasets.

Model	Precision	Recall	AUC
DeepDTA	0.938	0.935	0.972
DeepConv-DTI	0.939	0.907	0.967
MolTrans	0.955	0.933	0.974
TransformerCPI	0.911	0.937	0.970
IMAEN	0.952	0.945	0.954

mechanism. The results of these experiments are shown in Table 6, which presents the binding affinity prediction performance using three different evaluation metrics on the Davis independent test set for each alternative IMAEN model.

-Generated feature extraction channel: We removed the local augmentation module at the drug processing stage, that is, only the original feature was used as input.

-ISCEM: We removed the ISCEM at the protein processing stage and replaced it with the baseline target sequence processing module.

-Attention mechanism: To evaluate the contribution of the attention mechanism in the MSAM and ISCEM, we removed the attention mechanism.

From Table 6, we can obtain that generated feature extraction channel, ISCEM and attention mechanism together contribute to the model final performance. The ISCEM made a significant contribution to the final performance improvement. In addition, for the generated feature extraction channel, we see it is also helpful to increase the neighborhood information of molecules with fewer nodes by incorporating generated models for improved performance.

5.3. Parameter sensitivity analysis

In order to evaluate the impact of the number of layers in the GCN and CNN components of our proposed model and to find the right number of GCNs for the drug/protein processing module—neither too simple nor overly complex—in order to achieve optimal performance

Table 6
Effectiveness of different components of IMAEN (using Davis Dataset (pk_d)).

Setup	MSE	CI	r_m^2
IMAEN	0.211	0.905	0.705
-Generated feature extraction channel	0.223	0.896	0.693
-ISCEM	0.232	0.895	0.680
-Attention	0.221	0.896	0.685

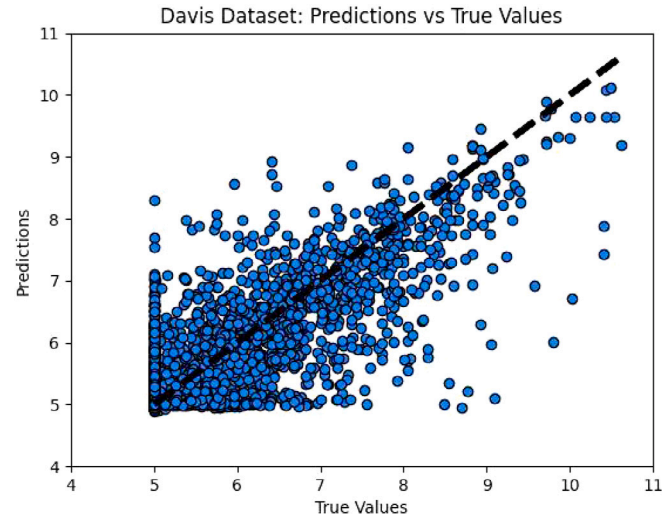


Fig. 4. IMAEN predictions against the binding affinity of predicted and actual value for Davis.

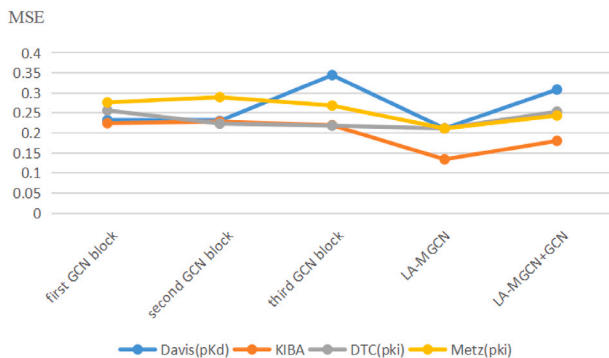


Fig. 5. The influence of GCN layer number variation in MSAM.

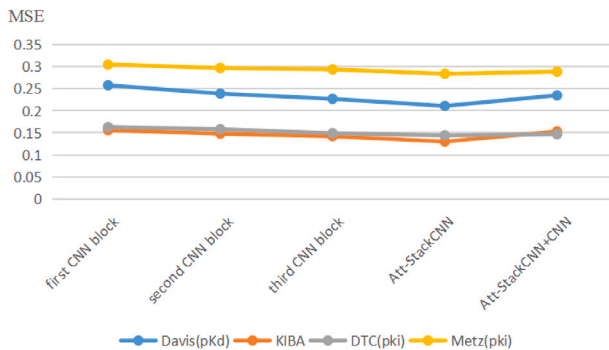


Fig. 6. The influence of CNN layer number variation in ISCEM.

and effectiveness when handling drug information, we conducted sensitivity analyses on the four datasets by varying the number of layers in the MSAM. These experiments were conducted to demonstrate the robustness of our approach. The experimental results are shown in Fig. 5. Each point on the curve represents a different combination of the number of GCN layers. Firstly, we enter the adjacency matrix A of the drug molecules into the first GCN block of MSAM and remove others. Then we input only the squared representation of A (i.e., A^2) into the second GCN block of MSAM and remove others. Finally, we input the cubic representation of A (i.e., A^3) into the third GCN block of MSAM and remove others. In accordance with the aforementioned setup, we trained three distinct models and recorded their respective MSE. The outcome of our experiment demonstrates that the combination of all three modules performs the best on all four datasets.

At the same time, we verify the influence of different CNN combinations in the ISCEM on the four datasets. As shown in Fig. 6, we input the embedded result of the target sequence into the first CNN block of ISCEM and remove others. Then we pass the embedded result of the target sequence as input into the first and second GCN block of ISCEM and remove others. Finally, we take the embedded result of the target sequence as input into the third CNN block of ISCEM. By using the previously mentioned setup, we train three different models and note their corresponding MSE. The results show that the combination of the three module achieves the best performance.

5.4. Visualization of attention mechanism

In this section, we delve deeper into the analysis of the attention weights learned by the models, aiming to gain a more comprehensive understanding of the mechanisms on which the models base their predictions. The visualization of attention weights is utilized to demonstrate the interpretability of the model. This is done to intuitively showcase how the model focuses on different features of protein sequences during the prediction process, particularly the interaction between drug molecules and protein sequences. This visualization aids in explaining the basis for the model's decisions and its prediction outcomes. While ISCEM is used in the protein feature extraction phase, MSAM is applied in the drug feature extraction phase. The utilization of these two modules does indeed contribute to explaining the model's focus on protein and drug features. To some extent, they can represent the interpretability of the entire model. As illustrated in Fig. 7, the left side displays the 3D protein structures, with the binding sites highlighted in red. On the right side, the amino acid sequence of the protein is shown, accompanied by the heatmap representing the distribution of attention weights. We have chosen OMP decarboxylase (PDB:1DBT) and its corresponding active sites as illustrative examples. In this visual representation, the redder residues indicate higher importance in the predictions, and the actual binding sites in the amino acid sequence are marked with deep red circles.

As depicted in the graph on the right side of Fig. 7, the heatmap reveals that sequence fragments with higher attention weights cover multiple positions, such as the 10th–12th, 60th–70th, 121st–123rd, 184th–186th, 193rd–195th, and 214th–216th positions in the amino acid sequence. The results obtained from IMAEN affirm that our model effectively localizes ligand binding sites to high-weight regions. These findings strongly indicate that our model can serve as a valuable tool for researchers to narrow down the search space for binding sites, thereby potentially expediting the drug discovery process.

Through these in-depth analyses, we not only gain a more holistic understanding of the model's performance but also garner essential insights into the mechanisms of protein–drug interactions. By delving into the attention weights learned by the model, we enhance our comprehension of how the model leverages various features for predicting drug–protein interactions. This understanding can guide our future research endeavors and model design, thereby advancing the field of drug discovery.

Table 7
The predicted candidate targets for new drug Diacerein.

Rank	Target name	Target Uniprot ID	Affinity	Evidence
1	Cytochrome P450 2C9	P11712	7.915	Tang et al. (2009)
2	Receptor-type tyrosine-protein kinase FLT3	P36888	7.908	Unknown
3	Polyunsaturated fatty acid 5-lipoxygenase	P09917	7.845	Singh et al. (2012)
4	Janus kinase 3 (JAK-3)	P52333	7.830	Unknown
5	Oxysterols receptor LXR-beta	P55055	7.825	Sheng et al. (2012)
6	Cholesterol 25-hydroxylase	P10635	7.765	Tang et al. (2009)
7	Tropomyosin-related kinase A	P04629	7.738	Unknown
8	Oxysterols receptor LXR-alpha	Q13133	7.709	Sheng et al. (2012)
9	Serine/threonine-protein kinase N1	Q16512	7.684	Unknown
10	CD140 antigen-like family member B	P09619	7.591	Unknown

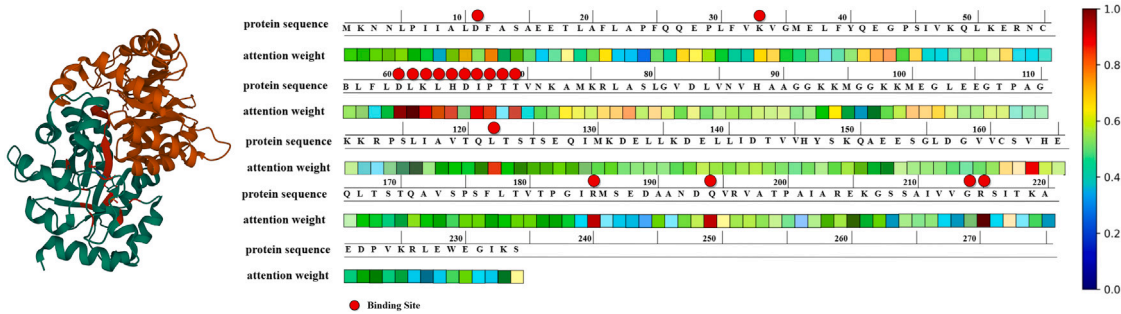


Fig. 7. Visualizations of attention weights of OMP decarboxylase (PDB:1DBT).

Table 8
The predicted candidate drugs for new target Aspartate aminotransferase.

Rank	Drug name	DrugBank ID	Affinity	Evidence
1	Aspartic acid	DB00128	7.093	Birsoy et al. (2015)
2	Dopamine	DB00988	7.845	Unknown
3	Dobutamine	DB00841	7.079	Unknown
4	Pyridoxal phosphate	DB00114	7.041	Tang et al. (2009)
5	Dobutamine	DB00841	7.017	Unknown
6	Voxelotor	DB14975	6.994	Unknown
7	Glutamic acid	DB00142	6.977	Toney (2014)
8	Resveratrol	DB02709	6.948	Unknown
9	Piceatannol	DB08399	6.921	Unknown
10	Darifenacin	DB00496	6.900	Unknown

5.5. Case study

To prove the effectiveness of our proposed model, we used IMAEN to predict new drugs [Diacerein (DrugBank ID: DB11994)] and new targets [Aspartate aminotransferase (Uniprot ID: Q2TU84)]. We can get the affinity score of known drugs and targets in the datasets by using the pretrained model. The predicted candidates were then ranked based on their affinity scores, and their interactions were subsequently validated on the DrugBank datasets.

Table 7 shows the top 10 Diacerein-related targets predicted by IMAEN. It can be seen from the table that 5 of the 10 targets were successfully predicted (marked in bold). For the top-ranked target, Cytochrome P450 2C9, we can get support from Tang et al. (2009). Similarly, for the top ten ranked Aspartate amino Transferase-related drug candidates predicted by IMAEN in Table 8, we predicted three drugs successfully (marked in bold). For the top-ranked drug, Aspartic acid, we can get support from Birsoy et al. (2015).

In general, predicting candidate drugs and targets for novel targets and drugs such as Diacerein and aspartate aminotransferase can be challenging. However, the results demonstrate that IMAEN can accurately predict the interaction between unknown drugs and target drug candidates from a large sample dataset. This is highly significant for drug screening and drug repositioning.

6. Conclusion

In this study, we propose an interpretable molecular augmentation encoding networks named IMAEN to predict drug–target interactions based on molecular structure augmentation module(MSAM) and an interpretable stack convolutional encoding module(ISCEN). IMAEN is designed to learn the node feature matrices of drugs and the embedding representations of proteins, which are then fed into the MSAM and ISCEN which adopt molecular structure augmentation mechanism and attention interpretability mechanism, respectively. The output representations are concatenated and passed through two fully connected layers to predict the DTI scores. Our experiments demonstrate that our method outperforms several state-of-the-art models, and the fusion of two mechanism is effective for DTI prediction. Ablation experiments show that the generated feature extraction channel, ISCEN and attention mechanism are useful and complementary, and their combination improves the DTI prediction results. Additionally, our visual experiments demonstrate that the attention mechanism in our model correctly learns the relationship between drug–protein pairs, providing interpretability for the model.

CRediT authorship contribution statement

Jing Zhang: Designed the method and experiments, Performed the experiments and analyzed the results, Wrote the manuscript. **Zhi**

Liu: Wrote the manuscript. **Yaohua Pan:** Performed the experiments and analyzed the results. **Hongfei Lin:** Provided suggestions and feedback. **Yijia Zhang:** Designed the method and experiments, Wrote the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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